What does heterogeneity cost? A controlled, communication-priced decomposition of federated LSTM load forecasting on London smart meters

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Abstract

Placeholder.

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1. Introduction

Short-term load forecasts at the level of a single household feed demand response, home energy management, local flexibility markets and the operation of low-voltage networks that now host heat pumps, electric vehicles and rooftop photovoltaics. Deep recurrent models, in particular the long short-term memory (LSTM) network [1], forecast individual households better than per-home statistical models when they are trained on many households at once [2]. Pooling the readings in one place, however, is what smart-meter regulation and customer consent increasingly discourage: half-hourly consumption reveals occupancy, routines and appliance use. Federated learning (FL) [3, 4] offers an alternative. Each meter, home gateway or community energy hub trains on its own data, and only model weights travel to an aggregator.

A large body of work now applies FL to residential load forecasting [5, 6, 7, 8, 9, 10, 11, 12, 13, 14]. Two premises run through it. The first

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is that household data are strongly non-identically distributed (non-IID), so that plain federated averaging (FedAvg) suffers, and that remedies such as the proximal term of FedProx [15], clustering [7, 16] or personalisation [9, 8, 12] are needed. The second is that FL is the communication-light, privacy-preserving option. Both premises have been tested only in isolation. On LCL, Briggs et al. [17] found plain FedAvg worse than purely local models and recovered the loss with clustering and local fine-tuning. Taïk and Cherkaoui [18] and Savi and Olivadese [19] estimated large bandwidth savings over raw-data upload, whereas Briggs et al. [20] argued, without measurement, that weight vectors cost more bandwidth than half-hourly readings on constrained meter links. Studies differ in dataset, cohort, horizon, model, number of rounds and evaluation protocol, and almost none holds the households and the test set fixed while varying only how the data are split across clients. As a result, the field cannot say how much of the accuracy lost by FL is caused by heterogeneity, how much by splitting the data at all, and how much by the federated training protocol itself. Nor can it say whether the bytes exchanged during training are small compared with simply uploading the raw readings.

This paper answers these questions with a controlled experiment on the Low Carbon London (LCL) smart-meter trial [21]. We fix a stratified cohort of 100 standard-tariff London households, one chronological train/validation/test split, one LSTM of 17217 parameters and one training budget. We then change exactly one factor at a time: whether data are pooled, the number of clients the same homes are split into, how homes are grouped into clients (from random to fully clustered by load profile), client participation and stragglers, the aggregation rule, and the precision of the transmitted update. Every comparison uses the same test half-hours, so differences in error are attributable to the manipulated factor alone.

Contributions.

1. **A controlled decomposition protocol for federated load forecasting.** We separate the gap between federated and centralised training into the cost of the round protocol (one client holding all data), the cost of fragmentation (the same homes split over K clients), the cost of heterogeneity (a Dirichlet dial over real load-profile clusters) and the cost of partial participation. Heterogeneity is reported as a *measured* index H between clients, computed on training data only, rather than as a nominal partition label.

2. **Communication-priced accuracy.** Prior estimates of FL's bandwidth advantage are analytic, derived from model and data sizes [18], or summarised as an overall ratio of exchanged data [19]. We count every byte uploaded and downloaded and introduce two budget metrics, R@ε and B@ε: the rounds and megabytes a federation needs to come within ε of centralised validation error. We compare these with the volume of the raw training data itself.
3. **Equity of federation.** We report, household by household, who gains and who loses relative to training alone, by consumption quintile, with paired signed-rank tests [22] and Holm correction [23]. We also evaluate a zero-communication personalisation step, one epoch of local fine-tuning of the global model.
4. **An update-dissimilarity schedule for the proximal coefficient, and why it does not help here.** Adaptive proximal coefficients exist, driven by the training loss [15] or by a local-versus-global performance index [24]. HA-FedProx instead schedules μ from a parameter-free, bounded, per-round estimate of FedProx's own B-dissimilarity [15], computed in update space from messages the server already receives. It is closely related to gradient diversity [25]. We show that on household load data this signal tracks training progress rather than data heterogeneity, and we explain why. This negative result applies to any scheme that adapts regularisation to observed update dissimilarity.
5. **An open, reproducible pipeline.** Preprocessing, partitioning, all arms, byte accounting, statistics, figures and the tables of this paper are generated by one public code base from the public LCL release. A Flower backend [26] reproduces the simulator exactly.

Section 2 reviews related work. Section 3 defines the task, the arms, the heterogeneity index, HA-FedProx and the communication metrics. Section 4 describes data and protocol. Section 5 reports the results, Section 6 discusses their implications for deployment, and Section 7 concludes.

2. Related work

2.1. Household load forecasting with deep learning

Household load is volatile and dominated by occupant behaviour. Kong et al. [2] showed that LSTMs outperform shallow and statistical baselines for individual residential customers. Shi et al. [27] showed that pooling the

data of neighbouring households reduces over-fitting, the effect that motivates collaborative training in the first place. Later work added embeddings, bidirectional recurrences and multi-task sharing [28, 29], as well as clustering of consumption profiles [30]. These gains assume the data are pooled.

2.2. Federated household load forecasting

FL for residential forecasting has grown quickly [5, 6, 7, 31, 10, 11, 9, 32, 33, 8, 16, 34, 35, 36, 37, 38, 39, 40, 41, 42, 12, 13, 14, 43, 44, 45, 46, 47, 48]. The findings that matter here disagree about whether federation pays off at all.

- *Federation can lose to local models.* The closest study to ours is Briggs et al. [17], who trained a two-layer LSTM with FedAvg on 100 LCL households. Plain FL was 7.6 % worse in mean RMSE than localised learning, clustering reduced the shortfall to 4.7 %, and local fine-tuning made the federated models better than local ones. Gholizadeh and Musilek [7] likewise found local learning best on average for individual Canadian homes.

- *Federation can match or beat local models.* Fekri et al. [6] found FedAvg comparable to or better than per-meter models on 19 Ontario homes.

- *Remedies for heterogeneity.* Personalised and clustered variants [9, 8, 16, 12, 13, 44] are motivated by heterogeneity among households.

None of these studies varies heterogeneity while holding the households and the test set fixed, so the share of the federated gap caused by heterogeneity, rather than by data fragmentation or the training protocol, has not been measured. We revisit the LCL comparison of Briggs et al. under a controlled protocol, with per-household paired statistics, and add that decomposition.

2.3. Communication cost

Communication efficiency was a founding motivation of FL [3, 49]. In load forecasting, the evidence points in both directions:

- Taïk and Cherkaoui [18] estimated analytically that FL saves 90–97 % of the network load of uploading data.

- Savi and Olivadese [19] reported up to 50 times less exchanged data at sub-hourly resolution.

- Briggs et al. [20] argued, without measurement, that exchanging weight vectors costs more bandwidth than sending half-hourly readings over constrained smart-meter links.

Most FL forecasting studies count rounds rather than bytes [6], or report payload per round without comparing it with the data [50]. We measure upload and download bytes per round, the bytes needed to reach a given accuracy, and the raw-data volume they replace.

2.4. Non-IID data and proximal regularisation

Statistical heterogeneity slows and biases FedAvg [51, 52, 4]. Remedies include:

- proximal regularisation (FedProx [15]);
- control variates (SCAFFOLD [53]);
- dynamic regularisation (FedDyn [54]);
- normalised averaging (FedNova [55]);
- adaptive server optimisers [56];
- personalisation [57, 58];
- fairness-aware objectives [59].

Heterogeneity is usually induced with a Dirichlet split over class labels [60]. Regression on smart-meter data has no class labels, so we use a Dirichlet split over load-profile clusters and also measure the resulting heterogeneity directly.

Choosing the proximal coefficient remains ad hoc. The FedProx paper adapts μ from the training loss [15]. HAPI-FedProx adapts the proximal term from a local-versus-global performance index [24]. Yu et al. [61] characterise the optimal regularisation strength as a function of heterogeneity in theory. FedAdp [62] uses the geometry of client updates to set aggregation weights. The update-disagreement ratio used by HA-FedProx is a bounded, update-space estimate of FedProx's B-dissimilarity, and it belongs to the same family as gradient diversity [25]. To our knowledge it has not been used to schedule μ, and its behaviour on real energy data has not been examined.

2.5. Privacy

FL keeps raw data local, but updates can leak training data through gradient inversion [63, 64]. We therefore claim data locality only, not formal privacy.

3. Methodology

3.1. Forecasting task and model

Let $x_{i,t}$ be the energy (kWh) consumed by household i in half-hour t. Each household's series is standardised with a mean and standard deviation estimated on its own training period only. The forecaster receives the last $L = 48$ standardised readings (24 h) and predicts the next half-hour, $\hat{x}_{i,t+1} = f_w(x_{i,t-L+1}, \ldots, x_{i,t})$. All arms share one architecture: a single-layer LSTM with 64 hidden units followed by a linear output layer ($P = 17217$ parameters, 68.9 kB in 32-bit floating point). Training minimises the mean squared error with Adam (learning rate 10^{-3}, batch 256). Errors are reported in kWh after inverting each household's scaling. Keeping the model small and fixed is deliberate: the object of study is the training regime, not the architecture.

3.2. Arms

Naive baselines. Persistence ($\hat{x}_{t+1} = x_t$) and seasonal naive ($\hat{x}_{t+1} = x_{t-47}$).

Local-only. One model per household, trained on that household's windows only, with early stopping on its own validation error. No data or weights leave the home.

Centralised. One model trained on the pooled windows of all households, with early stopping on pooled validation error. This is the accuracy reference, and it requires the raw readings to leave the home.

FedAvg. K clients each hold the windows of one or more households. In round t the server samples a set S_t of clients, sends them the global weights w_t, each client trains for E local epochs and returns w_k, and the server sets $w_{t+1} = \sum_{k \in S'_t} p_k w_k$ with $p_k = n_k / \sum_{j \in S'_t} n_j$, where n_k is the number of training windows of client k and $S'_t \subseteq S_t$ the clients that returned [3].

FedProx. Each client minimises $F_k(w) + \frac{\mu}{2}\|w - w_t\|^2$ and stragglers' partial updates are kept rather than dropped [15].

HA-FedProx. FedProx with a coefficient μ_t set by the server each round (Section 3.5).

FedAvg + local fine-tuning (FT). After federated training, each household fine-tunes the final global model for one epoch on its own training windows. This costs no communication and is evaluated for every federated run.

Clients are stateless: a fresh optimiser is created at every local update, as is standard for cross-device FL.

3.3. Clients and controlled partitions

Two client models are considered. In *household mode* each home is a client ($K = 100$), which reflects meter-level or home-gateway deployment and has natural, uncontrolled heterogeneity. In *community mode* the same homes are grouped into K clients, which reflects a building gateway, a community energy hub or an aggregator.

To dial heterogeneity with real, not invented, labels, every household is described by its normalised 24-bin mean daily load shape and its log mean load, both computed on the training period. These features are standardised and clustered with k-means ($k = 5$, 10 restarts) into load-profile groups such as evening-peaking or flat, low or high consumption. Households are then assigned to clients by one of three strategies: (i) *random*, where homes are shuffled into equal groups; (ii) *Dirichlet*(α) [60], where each client draws mixture proportions $\pi_k \sim \mathrm{Dir}(\alpha\mathbf{1})$ over profile clusters and fills its slots by sampling a cluster from π_k, renormalised to clusters with homes left; and (iii) *cluster*, where homes are sorted by cluster and mean load and chunked, the maximally separated case. Because the homes, and therefore the test windows, are identical under every strategy, error differences between partitions are caused by the partition alone.

3.4. Measured heterogeneity index

Nominal labels such as "$\alpha = 0.3$" do not say how different the resulting clients are. We therefore measure heterogeneity directly on training data. For client k, let $q_k \in \Delta^{167}$ be the distribution of its total training consumption

over the 168 hours of the week. With JS the Jensen–Shannon divergence in base 2 [65] and m_k the mean load of client k,

$$H_{\mathrm{JS}} = \binom{K}{2}^{-1} \sum_{k<l} \mathrm{JS}(q_k \,\|\, q_l), \qquad H_{\mathrm{CV}} = \frac{\mathrm{sd}(m_k)}{\mathrm{mean}(m_k)}, \qquad H = H_{\mathrm{JS}} + \frac{H_{\mathrm{CV}}}{1 + H_{\mathrm{CV}}}. \tag{1}$$

$H_{\mathrm{JS}} \in [0, 1]$ measures differences in *when* clients consume, and the bounded transform of H_{CV} measures differences in *how much* they consume. $H = 0$ for a single client or identical clients. The index uses no labels and no test data.

3.5. HA-FedProx: a heterogeneity-adaptive proximal coefficient

The proximal coefficient of FedProx trades client drift against local progress. It is usually fixed by tuning, yet the appropriate strength should depend on how much the client updates disagree. After aggregating round t, the server holds the client updates $\Delta_k = w_k - w_t$ and their weighted mean $\bar{\Delta} = \sum_k p_k \Delta_k$. It computes the *update-disagreement ratio*

$$\rho_t = \frac{\sum_k p_k \|\Delta_k - \bar{\Delta}\|^2}{\sum_k p_k \|\Delta_k\|^2} \in [0, 1], \tag{2}$$

which is 0 when all clients move identically and approaches 1 when their updates cancel. By the bias–variance identity $\sum_k p_k \|\Delta_k\|^2 = \|\bar{\Delta}\|^2 + \sum_k p_k \|\Delta_k - \bar{\Delta}\|^2$, $\rho_t = 1 - 1/B_t^2$ with $B_t^2 = \sum_k p_k \|\Delta_k\|^2 / \|\bar{\Delta}\|^2$, an update-space analogue of the B-local dissimilarity used in the convergence analysis of FedProx [15]. The next round uses

$$\mu_{t+1} = \mathrm{clip}(\mu_0 \, \rho_t, \ \mu_{\min}, \ \mu_{\max}) \,, \tag{3}$$

with $\mu_{\min} = 10^{-4}$, $\mu_{\max} = 1$ and μ_0 tuned on validation data exactly as FedProx's fixed μ. The rule needs no additional client computation or messages, and it reduces to FedAvg for $\mu_0 = 0$. An earlier variant normalised by $\|\bar{\Delta}\|^2$ alone; it is unbounded as the global model converges and $\|\bar{\Delta}\| \to 0$, which is why the bounded ratio (2) is used.

3.6. Partial participation and stragglers

In each round a fraction C of clients is sampled. Each sampled client independently becomes a straggler with probability p_s and completes only a uniform random fraction $u \sim U(0.1, 1)$ of its local work. Following the protocol of Li et al. [15], FedAvg drops stragglers (they download but never upload) while FedProx and HA-FedProx aggregate their partial updates.

3.7. Communication accounting

A model message carries $P \cdot b$ bytes, with $b = 4$ for 32-bit and $b = 2$ for 16-bit updates. Round t costs $Pb(|S_t| + |S'_t|)$: one download per selected client and one upload per returning client. Protocol framing is excluded; the Flower backend measures it at about 1 % of the payload. For the centralised arm we report the volume of the raw training readings, at 4 bytes per value plus an 8-byte timestamp, as the bytes that pooling would move. Let v_t be the validation MAE of the global model after round t and $v^\star$ the centralised validation MAE. We define

$$R@\varepsilon = \min\{t : v_t \le (1+\varepsilon)v^\star\}, \qquad B@\varepsilon = \sum_{s \le R@\varepsilon} Pb(|S_s| + |S'_s|), \tag{4}$$

with $\varepsilon = 10\,\%$. These convert "how close to centralised" into "at what network price".

3.8. Metrics and statistics

We report mean absolute error (MAE), root mean squared error (RMSE), weighted absolute percentage error (WAPE $= 100 \sum |x - \hat{x}| / \sum |x|$), and MAPE restricted to half-hours with $x \ge 0.05\,\text{kWh}$. Unrestricted MAPE is unbounded at the near-zero night-time loads that are common in households. All pooled metrics are micro-averaged over every test half-hour of every household. Per-household errors are averaged over seeds and compared with two-sided Wilcoxon signed-rank tests [22] over the 100 households, with Holm correction [23] across the family of comparisons. Because p-values with 100 paired households can be small for negligible effects, every test is reported with its median paired difference and the share of households for which one arm is better.

4. Experimental setup

4.1. Data

We use the public release of the Low Carbon London smart-meter trial run by UK Power Networks [21]. It contains half-hourly consumption of 5,567 London households between November 2011 and February 2014. We stream all 168 files of the partitioned release (93.1 million readings in the study year), observe 5,528 household identifiers, and apply the following rules. The study period is calendar year 2013. Only standard-tariff households are kept, because the time-of-use group received dynamic price signals

in 2013 that would confound the comparison. Unparseable, negative or implausible readings (> 10 kWh per half-hour) are set to missing, duplicate timestamps are averaged, and each series is re-indexed to a full 30-minute grid. Gaps of up to 2 h are linearly interpolated and longer gaps are left missing; any window that touches a missing value is dropped. Households with at least 95 % coverage in 2013 are eligible (4,034 homes). From these we draw a cohort of 100 homes, stratified into quintiles of training-period mean load (20 per quintile) with a fixed selection seed. The cohort's mean training-period load is 0.206 kWh per half-hour (range 0.008–0.972), and its mean coverage is 99.9 %.

The split is chronological and identical for every household: training January–August 2013, validation September–October and test November–December. A window belongs to the split of its target time stamp, so its inputs may reach back into the previous split; those readings are known at forecast time. This yields 1,160,952 training and 292,052 test windows. Scalers, load-profile clusters and the heterogeneity index are computed from training data only, and every decision (early stopping, best federated round, hyper-parameters) uses validation error only. Test error never drives a decision.

4.2. Experimental matrix

Table 1 lists the experiments. Unless stated otherwise, community mode uses $K = 20$ clients of five homes each, every client participates in every round, $E = 1$ local epoch and $T = 30$ rounds. Thirty rounds of one local epoch equal the number of passes over the data allowed to the centralised arm (30 epochs, early stopping with patience 4), so federated and centralised training have the same computational budget. FedProx's μ and HA-FedProx's μ_0 were tuned on validation data from the grids $\{10^{-3}, 10^{-2}, 10^{-1}\}$ and $\{10^{-2}, 10^{-1}, 1\}$, under the hardest (cluster) partition and with the first seed. This selected $\mu = 0.001$ and $\mu_0 = 0.01$, which were then used in every other experiment. Each experiment is repeated with three seeds (42, 43, 44), which change model initialisation, client sampling, stragglers and the random and Dirichlet groupings, but not the cohort, the profile clusters or the split.

4.3. Implementation

The pipeline is written in Python with PyTorch. The federated loop is a deterministic in-process simulator whose clients implement the Flower `NumPyClient` interface. An optional backend runs the same clients through

Table 1: Experimental matrix. All rows use the same 100 households, split, model and budget.

ID	Arm(s)	Factor varied
E0	persistence, seasonal naive	reference
E1	local-only	no collaboration
E2	centralised	pooled raw data
E3	FedAvg	random grouping, $K = 20$
E4, E5	FedAvg, FedProx	Dirichlet $\alpha = 0.3$
E6, E7	FedAvg, FedProx	cluster grouping (strong non-IID)
E8, E9, E11	FedAvg, FedProx, HA-FedProx	$\alpha = 0.3$, $C = 0.5$, 20 % stragglers
E10	FedAvg, FedProx, HA-FedProx	$\alpha \in \{\infty, 1, 0.3, 0.1, 0\}$ sweep
E12	FedAvg	$\alpha = 0.3$, 16-bit updates
E13	FedAvg, FedProx, HA-FedProx	one household per client ($K = 100$)
E14	FedAvg	random grouping, $K \in \{1, 5, 50\}$

Flower's own FedAvg and FedProx strategies [26], and on a test configuration it reproduces the simulator's per-round validation error and final test error exactly. Experiments ran on a single Apple M5 Max workstation (Metal backend), eight runs in parallel. Code, configuration files and every result table are available in the accompanying repository (Section 7).

5. Results

6. Discussion

7. Conclusion

Data and code availability

The Low Carbon London smart-meter data are publicly available from the London Datastore (`https://data.london.gov.uk/dataset/smartmeter-energy-consumption-data-in-london-households-vqm0d`) and are not redistributed by the authors. All code, configuration files, per-run results (summaries, per-round histories and per-household errors) and the scripts that generate every table and figure in this paper are available at [repository URL to be inserted on acceptance; provided to reviewers as a supplementary archive].

CRediT authorship contribution statement

Pushkal Gupta: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft. **Kaushik Vanka:** Conceptualization, Investigation, Validation, Writing – review & editing.

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References

[1] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Computation 9 (8) (1997) 1735–1780. doi:10.1162/neco.1997.9.8.1735.

[2] W. Kong, Z. Y. Dong, Y. Jia, D. J. Hill, Y. Xu, Y. Zhang, Short-term residential load forecasting based on LSTM recurrent neural network, IEEE Transactions on Smart Grid 10 (1) (2019) 841–851. doi:10.1109/TSG.2017.2753802.

[3] B. McMahan, E. Moore, D. Ramage, S. Hampson, B. Agüera y Arcas, Communication-efficient learning of deep networks from decentralized data, in: Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), Vol. 54 of Proceedings of Machine Learning Research, 2017, pp. 1273–1282.
URL https://proceedings.mlr.press/v54/mcmahan17a.html

[4] P. Kairouz, H. B. McMahan, et al., Advances and open problems in federated learning, Foundations and Trends in Machine Learning 14 (1–2) (2021) 1–210. doi:10.1561/2200000083.

[5] J. D. Fernández, S. P. Menci, C. M. Lee, A. Rieger, G. Fridgen, Privacy-preserving federated learning for residential short-term load forecasting, Applied Energy 326 (2022) 119915. doi:10.1016/j.apenergy.2022.119915.

[6] M. N. Fekri, K. Grolinger, S. Mir, Distributed load forecasting using smart meter data: Federated learning with recurrent neural networks, International Journal of Electrical Power & Energy Systems 137 (2022) 107669. doi:10.1016/j.ijepes.2021.107669.

[7] N. Gholizadeh, P. Musilek, Federated learning with hyperparameter-based clustering for electrical load forecasting, Internet of Things 17 (2022) 100470. doi:10.1016/j.iot.2021.100470.

[8] F. Widmer, S. Nowak, B. Bowler, P. Huber, A. Papaemmanouil, Data-driven comparison of federated learning and model personalization for electric load forecasting, Energy and AI 14 (2023) 100253. doi:10.1016/j.egyai.2023.100253.

[9] X. Qu, C. Guan, G. Xie, Z. Tian, K. Sood, C. Sun, L. Cui, Personalized federated learning for heterogeneous residential load forecasting, Big Data Mining and Analytics 6 (4) (2023) 421–432. doi:10.26599/BDMA.2022.9020043.

[10] Y. Liu, Z. Dong, B. Liu, Y. Xu, Z. Ding, FedForecast: A federated learning framework for short-term probabilistic individual load forecasting in smart grid, International Journal of Electrical Power & Energy Systems 152 (2023) 109172. doi:10.1016/j.ijepes.2023.109172.

[11] M. N. Fekri, K. Grolinger, S. Mir, Asynchronous adaptive federated learning for distributed load forecasting with smart meter data, International Journal of Electrical Power & Energy Systems 153 (2023) 109285. doi:10.1016/j.ijepes.2023.109285.

[12] P. Das, S. Pandey, C. Pandey, D. Kumar, Heterogeneity-aware personalised federated learning for household energy forecasting on multi-source real-world smart meter data, Scientific Reports 16 (2026) 24957. doi:10.1038/s41598-026-53020-6.

[13] J. Wang, D. Xu, L. Xing, T. Chen, Y. Liu, M. Shahidehpour, T. Yang, Personalized federated learning with mixture of experts and conformal prediction for household energy forecasting, Expert Systems with Applications 300 (2026) 130417. doi:10.1016/j.eswa.2025.130417.

[14] A. Tibermacine, I. Naidji, I. E. Tibermacine, S. Sahraoui, M. Zouai, A. Rabehi, M. Habib, Privacy-preserving load forecasting in smart grids using federated learning: a comparative analysis of aggregation strategies, Frontiers in Energy Research 14 (2026) 1854654. doi:10.3389/fenrg.2026.1854654.

[15] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, V. Smith, Federated optimization in heterogeneous networks, in: Proceedings of Machine Learning and Systems (MLSys), Vol. 2, 2020, pp. 429–450. URL https://proceedings.mlsys.org/paper_files/paper/2020/hash/1f5fe83998a09396ebe6477d9475ba0c-Abstract.html

[16] A. Dogra, A. Anand, J. Bedi, Consumers profiling based federated learning approach for energy load forecasting, Sustainable Cities and Society 98 (2023) 104815. doi:10.1016/j.scs.2023.104815.

[17] C. Briggs, Z. Fan, P. Andras, Federated learning for short-term residential load forecasting, IEEE Open Access Journal of Power and Energy 9 (2022) 573–583. doi:10.1109/OAJPE.2022.3206220.

[18] A. Taïk, S. Cherkaoui, Electrical load forecasting using edge computing and federated learning, in: ICC 2020 – 2020 IEEE International Conference on Communications (ICC), 2020, pp. 1–6. doi:10.1109/ICC40277.2020.9148937.

[19] M. Savi, F. Olivadese, Short-term energy consumption forecasting at the edge: A federated learning approach, IEEE Access 9 (2021) 95949–95969. doi:10.1109/ACCESS.2021.3094089.

[20] C. Briggs, Z. Fan, P. Andras, Privacy preserving demand forecasting to encourage consumer acceptance of smart energy meters, in: NeurIPS 2020 Workshop on Tackling Climate Change with Machine Learning, 2020. arXiv:2012.07449, doi:10.48550/arXiv.2012.07449.

[21] UK Power Networks, SmartMeter energy consumption data in London households (low carbon london project), London Datastore, https://data.london.gov.uk/dataset/smartmeter-energy-consumption-data-in-london-households-vqm0d, cC BY 4.0. 5,567 households, half-hourly kWh, Nov 2011–Feb 2014. Accessed 8 October 2026. Release year not stated on the Datastore page (2015).

[22] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics Bulletin 1 (6) (1945) 80–83. doi:10.2307/3001968.

[23] S. Holm, A simple sequentially rejective multiple test procedure, Scandinavian Journal of Statistics 6 (2) (1979) 65–70, no DOI registered; JSTOR stable URL given.
URL https://www.jstor.org/stable/4615733

[24] K. Mhou, M. Senekane, HAPI-FedProx: Heterogeneity-aware adaptive proximal optimization for federated learning, in: A. Gerber, A. W. Pillay (Eds.), Artificial Intelligence Research (SACAIR), Communications in Computer and Information Science, Springer Nature Switzerland, 2025, pp. 272–286, cCIS volume number not verified; online 25 Nov 2025, volume dated 2026. doi:10.1007/978-3-032-11733-5_17.

[25] D. Yin, A. Pananjady, M. Lam, D. Papailiopoulos, K. Ramchandran, P. Bartlett, Gradient diversity: a key ingredient for scalable distributed learning, in: Proceedings of the 21st International Conference on Artificial Intelligence and Statistics (AISTATS), Vol. 84 of Proceedings of

Machine Learning Research, 2018, pp. 1998–2007.
URL https://proceedings.mlr.press/v84/yin18a.html

[26] D. J. Beutel, T. Topal, A. Mathur, X. Qiu, J. Fernandez-Marques, Y. Gao, L. Sani, K. H. Li, T. Parcollet, P. P. B. de Gusmão, N. D. Lane, Flower: A friendly federated learning research framework, arXiv preprint arXiv:2007.14390 (2020). doi:10.48550/arXiv.2007.14390.

[27] H. Shi, M. Xu, R. Li, Deep learning for household load forecasting—a novel pooling deep RNN, IEEE Transactions on Smart Grid 9 (5) (2018) 5271–5280. doi:10.1109/TSG.2017.2686012.

[28] K. Aurangzeb, S. I. Haider, M. Alhussein, Individual household load forecasting using bi-directional LSTM network with time-based embedding, Energy Reports 11 (2024) 3963–3975. doi:10.1016/j.egyr.2024.03.028.

[29] J.-W. Xiao, M. Cao, H. Fang, J. Wang, Y.-W. Wang, Joint load prediction of multiple buildings using multi-task learning with selected-shared-private mechanism, Energy and Buildings 293 (2023) 113178. doi:10.1016/j.enbuild.2023.113178.

[30] K. Dab, N. Henao, S. Nagarsheth, Y. Dubé, S. Sansregret, K. Agbossou, Consensus-based time-series clustering approach to short-term load forecasting for residential electricity demand, Energy and Buildings 299 (2023) 113550. doi:10.1016/j.enbuild.2023.113550.

[31] Y. Yang, Z. Wang, S. Zhao, J. Wu, An integrated federated learning algorithm for short-term load forecasting, Electric Power Systems Research 214 (2023) 108830. doi:10.1016/j.epsr.2022.108830.

[32] R. Wang, H. Yun, R. Rayhana, J. Bin, C. Zhang, O. E. Herrera, Z. Liu, W. Mérida, An adaptive federated learning system for community building energy load forecasting and anomaly prediction, Energy and Buildings 295 (2023) 113215. doi:10.1016/j.enbuild.2023.113215.

[33] M. Khalil, M. Esseghir, L. Merghem Boulahia, Privacy-preserving federated learning: An application for big data load forecast in buildings, Computers & Security 131 (2023) 103211. doi:10.1016/j.cose.2023.103211.

[34] E. M. de Moraes Sarmento, I. F. Ribeiro, P. R. N. Marciano, Y. G. Neris, H. R. d. O. Rocha, V. F. S. Mota, R. d. S. Villaça, Forecasting energy power consumption using federated learning in edge computing devices, Internet of Things 25 (2024) 101050. doi:10.1016/j.iot.2023.101050.

[35] N. Abdulla, M. Demirci, S. Ozdemir, Smart meter-based energy consumption forecasting for smart cities using adaptive federated learning, Sustainable Energy, Grids and Networks 38 (2024) 101342. doi:10.1016/j.segan.2024.101342.

[36] G. Singh, J. Bedi, A federated and transfer learning based approach for households load forecasting, Knowledge-Based Systems 299 (2024) 111967. doi:10.1016/j.knosys.2024.111967.

[37] H. U. Manzoor, A. Jafri, A. Zoha, Adaptive single-layer aggregation framework for energy-efficient and privacy-preserving load forecasting in heterogeneous federated smart grids, Internet of Things 28 (2024) 101376. doi:10.1016/j.iot.2024.101376.

[38] Q. Mao, L. Wang, Y. Long, L. Han, Z. Wang, K. Chen, A blockchain-based framework for federated learning with privacy preservation in power load forecasting, Knowledge-Based Systems 284 (2024) 111338. doi:10.1016/j.knosys.2023.111338.

[39] Y. Huang, Y. Song, Z. Jing, Load forecasting using federated learning with considering electricity data privacy preservation of EASP, Ain Shams Engineering Journal 15 (6) (2024) 102724. doi:10.1016/j.asej.2024.102724.

[40] M. Mostafanezhad, M. Mohammadi, D. Keihan Asl, H. Karimipour, Personalized poison attack tolerant federated learning for residential load forecasting, Sustainable Energy, Grids and Networks 43 (2025) 101794. doi:10.1016/j.segan.2025.101794.

[41] I. Kaur, J. Bedi, A. Aggarwal, Privacy preserving and computationally efficient federated learning approach for household load estimation, Sustainable Energy, Grids and Networks 44 (2025) 102021. doi:10.1016/j.segan.2025.102021.

[42] L. Shao, K. Pang, C. Feng, Y. Yang, F. Wen, Privacy-preserving residential short-term load forecasting with missing data tolerance, IEEE Transactions on Smart Grid 17 (2) (2026) 1664–1679. doi:10.1109/TSG.2025.3628876.

[43] Z. He, H. Zhang, Y. Liu, Privacy-preserving electricity load forecasting using hierarchical federated learning, Journal of Information and Intelligence 4 (4) (2026) 350–363. doi:10.1016/j.jiixd.2026.04.009.

[44] N. Lu, S. Liu, Q. Wen, Q. Chen, L. Sun, Y. Wang, Federated domain separation for distributed forecasting of non-IID household loads, IEEE Transactions on Smart Grid 15 (4) (2024) 4271–4283. doi:10.1109/TSG.2024.3367766.

[45] M. A. Husnoo, A. Anwar, N. Hosseinzadeh, S. N. Islam, A. N. Mahmood, R. Doss, A secure federated learning framework for residential short-term load forecasting, IEEE Transactions on Smart Grid 15 (2) (2024) 2044–2055. doi:10.1109/TSG.2023.3292382.

[46] Y. He, F. Luo, M. Sun, G. Ranzi, Privacy-preserving and hierarchically federated framework for short-term residential load forecasting, IEEE Transactions on Smart Grid 14 (6) (2023) 4409–4423. doi:10.1109/TSG.2023.3268633.

[47] A. Duttagupta, J. Zhao, S. Shreejith, Exploring lightweight federated learning for distributed load forecasting, in: 2023 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), 2023, pp. 1–6. doi:10.1109/SmartGridComm57358.2023.10333889.

[48] R. Rahman, P. Moriano, S. U. Khan, D. C. Nguyen, Electrical load forecasting over multihop smart metering networks with federated learning, IEEE Internet of Things Journal 12 (18) (2025) 38413–38426. doi:10.1109/JIOT.2025.3586115.

[49] J. Konečný, H. B. McMahan, F. X. Yu, P. Richtárik, A. T. Suresh, D. Bacon, Federated learning: Strategies for improving communication efficiency, arXiv preprint arXiv:1610.05492 (2016). doi:10.48550/arXiv.1610.05492.

[50] S. Bose, K. Kim, Federated short-term load forecasting with personalization layers for heterogeneous clients, arXiv preprint arXiv:2309.13194 (2023). doi:10.48550/arXiv.2309.13194.

[51] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, V. Chandra, Federated learning with non-IID data, arXiv preprint arXiv:1806.00582 (2018). doi:10.48550/arXiv.1806.00582.

[52] K. Hsieh, A. Phanishayee, O. Mutlu, P. Gibbons, The non-IID data quagmire of decentralized machine learning, in: Proceedings of the 37th International Conference on Machine Learning (ICML), Vol. 119 of Proceedings of Machine Learning Research, 2020, pp. 4387–4398.
URL https://proceedings.mlr.press/v119/hsieh20a.html

[53] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, A. T. Suresh, SCAFFOLD: Stochastic controlled averaging for federated learning, in: Proceedings of the 37th International Conference on Machine Learning (ICML), Vol. 119 of Proceedings of Machine Learning Research, 2020, pp. 5132–5143.
URL https://proceedings.mlr.press/v119/karimireddy20a.html

[54] D. A. E. Acar, Y. Zhao, R. Matas Navarro, M. Mattina, P. N. Whatmough, V. Saligrama, Federated learning based on dynamic regularization, in: International Conference on Learning Representations (ICLR), 2021.
URL https://arxiv.org/abs/2111.04263

[55] J. Wang, Q. Liu, H. Liang, G. Joshi, H. V. Poor, Tackling the objective inconsistency problem in heterogeneous federated optimization, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 33, 2020, pp. 7611–7623.
URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/564127c03caab942e503ee6f810f54fd-Abstract.html

[56] S. Reddi, Z. Charles, M. Zaheer, Z. Garrett, K. Rush, J. Konečný, S. Kumar, H. B. McMahan, Adaptive federated optimization, in: International Conference on Learning Representations (ICLR), 2021.
URL https://arxiv.org/abs/2003.00295

[57] A. Fallah, A. Mokhtari, A. Ozdaglar, Personalized federated learning with theoretical guarantees: A model-agnostic meta-learning approach, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 33, 2020, pp. 3557–3568.
URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/24389bfe4fe2eba8bf9aa9203a44cdad-Abstract.html

[58] T. Li, S. Hu, A. Beirami, V. Smith, Ditto: Fair and robust federated learning through personalization, in: Proceedings of the 38th International Conference on Machine Learning (ICML), Vol. 139 of Proceedings of Machine Learning Research, 2021, pp. 6357–6368.
URL https://proceedings.mlr.press/v139/li21h.html

[59] T. Li, M. Sanjabi, A. Beirami, V. Smith, Fair resource allocation in federated learning, in: International Conference on Learning Representations (ICLR), 2020.
URL https://arxiv.org/abs/1905.10497

[60] T.-M. H. Hsu, H. Qi, M. Brown, Measuring the effects of non-identical data distribution for federated visual classification, arXiv preprint arXiv:1909.06335 (2019). doi:10.48550/arXiv.1909.06335.

[61] X. Yu, Z. He, Y. Sun, L. Xue, R. Li, Understanding the statistical accuracy-communication trade-off in personalized federated learning with minimax guarantees, in: Proceedings of the 42nd International Conference on Machine Learning (ICML), Vol. 267 of Proceedings of Machine Learning Research, 2025, pp. 73009–73050.
URL https://arxiv.org/abs/2410.08934

[62] H. Wu, P. Wang, Fast-convergent federated learning with adaptive weighting, IEEE Transactions on Cognitive Communications and Networking 7 (4) (2021) 1078–1088. doi:10.1109/TCCN.2021.3084406.

[63] L. Zhu, Z. Liu, S. Han, Deep leakage from gradients, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 32, 2019.
URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/60a6c4002cc7b29142def8871531281a-Abstract.html

[64] J. Geiping, H. Bauermeister, H. Dröge, M. Moeller, Inverting gradients – how easy is it to break privacy in federated learning?, in: Advances

in Neural Information Processing Systems (NeurIPS), Vol. 33, 2020, pp. 16937–16947.
URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/c4ede56bbd98819ae6112b20ac6bf145-Abstract.html

[65] J. Lin, Divergence measures based on the Shannon entropy, IEEE Transactions on Information Theory 37 (1) (1991) 145–151. doi:10.1109/18.61115.